

RES-PHY — Tsunami and Coastal Fluid Dynamics

July 19, 2026

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1 Domain Purpose and Scope

2 Domain Deliverables

3 RES-PHY-SRC — Tsunami Generation and Source Physics

3.1 Purpose and Scope

3.2 Research Questions

3.3 Terminology and Definitions

3.4 Literature Evidence

3.4.1 Fujii et al. (2011)

Fujii et al. [1]: Fujii et al. reconstructed the tsunami source of the 2011 Tohoku earthquake through waveform inversion using coastal tide and wave gauges, offshore GPS wave gauges, ocean-bottom pressure sensors, and DART records¹. The source was divided into forty rectangular subfaults measuring approximately 50×50 km, with a common strike of 193° , dip of 14° , and rake of 81° . The inferred slip distribution was converted into static seabed deformation using an Okada-type elastic half-space model, with the influence of coseismic horizontal displacement included over steep bathymetric gradients.

The paper is retained as the first candidate source reconstruction for the Tohoku benchmark. Its principal contribution is an observationally constrained finite-fault slip distribution that can be converted into a seabed-deformation field. Its principal limitations are the coarse subfault resolution, substantial uncertainty in the largest inferred slips, simplified rupture kinematics, and the use of the same tsunami waveforms for both inversion and waveform comparison.

3.5 Governing Relations and Quantitative Definitions

3.6 Physical Mechanisms

Fujii et al. [1]: The reconstructed tsunami consisted of two physically distinct source components. Very large slip near the Japan Trench generated the shorter-period impulsive tsunami component, whereas deeper interplate slip beneath southern Sanriku-oki and Miyagi-oki produced the preceding gradual water-level in-

crease and contributed to the extensive inundation of the Sendai Plain. The source should therefore not be represented as a spatially uniform fault or an arbitrary Gaussian free-surface disturbance, as either simplification would remove the observed distinction between the impulsive near-trench and broader interplate components.

3.7 Characteristic Spatial and Temporal Scales

Fujii et al. [1]: The region with inferred slip exceeding 2 m extended for approximately 350 km. The two largest subfault slips exceeded 40 m, while the source model assumed a constant rise time of 30 s. Offshore observations contained an initial gradual increase in water level of approximately 2 m followed by an impulsive wave of approximately 3–5 m with a period of approximately 8 min.

3.8 Relevant Observables

3.9 Controlling Dimensionless Parameters

3.10 Physical Regimes and Transition Conditions

3.11 Implications for Model Validity

3.12 Candidate Approaches

Fujii et al. [1]: The finite-fault inversion is retained as Candidate Source Model A for the benchmark simulation. Under this approach, the reported subfault slips are converted into vertical and horizontal seabed displacement before the resulting deformation is transferred to the hydrodynamic model. The tabulated slip values are not themselves water-surface elevations and shall not be imposed directly as the tsunami initial condition.

3.13 Critical Comparison

Fujii et al. [1]: The principal strength of the source model is its direct constraint by near-field and far-field tsunami observations. Its principal weaknesses are the coarse 50 km subfault resolution, substantial uncertainty in the largest inferred slips, simplified rupture kinematics, and reliance on linear shallow-water propagation during the waveform inversion. Moreover, the observations used to infer the source cannot subsequently be treated as an independent validation dataset.

3.14 Assumptions and Applicability

Fujii et al. [1]: The model assumes static elastic seabed deformation, a constant 30 s rise time on every subfault, instantaneous rupture propagation, and uniform rigidity of 5.0×10^{10} Pa. The reported slip distribution is applicable as a first observationally constrained reconstruction of the large-scale tsunami source, but does not establish that dynamic rupture effects or smaller-scale source heterogeneity are negligible for the present project.

3.15 Limitations and Failure Conditions

Fujii et al. [1]: The source reconstruction did not fully explain the largest reported run-up values in central Sanriku, indicating that additional or more localised source components may be absent. The inferred slip errors were estimated through delete-half jackknife analysis and shall not be interpreted automatically as complete Gaussian uncertainty distributions.

3.16 Project-Specific Conclusions

Fujii et al. [1]: The Fujii et al. finite-fault model shall be retained as the first candidate earthquake source for the Tohoku benchmark. Acceptance as the final source model is deferred until comparison with a later or independently reconstructed source. The source-severity study should consider independent perturbation of the near-trench and deeper interplate slip components rather than relying solely on uniform scaling of the complete slip field.

3.17 Selected Approach and Rationale

Fujii et al. [1]: The source-data workstream shall preserve the subfault number, north-east-corner latitude and longitude, depth, slip, reported error, patch dimensions, strike, dip, rake, rise time, and rigidity. The mathematical-model workstream shall define how the fault slip is converted into seabed deformation and how that deformation is applied to the hydrodynamic model. The verification workstream shall distinguish reproduction of the inversion waveforms from independent validation against held-out observations.

For the regional NLSWE model, the selected source representation is a dynamic moving-bed field $b(x, y, t) = b_0(x, y) + \sum_m \Delta b_m(x, y) R_m(t)$. Static displacement can remain a verification/fallback case, but it is not the active source representation for the locked regional mathematical model. The exact fault-source dataset remains a RES-DAT-SRC decision.

3.18 Unresolved Decisions

3.19 Requirements and Handoffs

3.20 Deliverable Completion Record

4 RES-PHY-PRO — Offshore Propagation and Wave Transformation

4.1 Purpose and Scope

4.2 Research Questions

4.3 Terminology and Definitions

4.4 Literature Evidence

4.5 Governing Relations and Quantitative Definitions

4.6 Physical Mechanisms

4.7 Characteristic Spatial and Temporal Scales

4.8 Relevant Observables

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4.11 Implications for Model Validity

4.12 Candidate Approaches

4.13 Critical Comparison

4.14 Assumptions and Applicability

4.15 Limitations and Failure Conditions

4.16 Project-Specific Conclusions

The regional 2D NLSWE model is the established baseline for offshore-to-nearshore propagation in the initial specification. It must still be validated against selected Tohoku observations before propagation credibility is claimed for a case study.

4.17 Selected Approach and Rationale

4.18 Unresolved Decisions

4.19 Requirements and Handoffs

‘RES-VER-SPEC’ shall include offshore waveform, arrival-time, amplitude, phase and boundary-reflection checks. ‘RES-DAT-GEO’ shall provide bathymetry and domain extents sufficient to avoid artificial boundary contamination.

4.20 Deliverable Completion Record

5 RES-PHY-NRS — Nearshore Transformation, Run-Up and Inundation

5.1 Purpose and Scope

5.2 Research Questions

5.3 Terminology and Definitions

5.4 Literature Evidence

5.4.1 Mori et al. (2012)

Mori, Takahashi, and The 2011 Tohoku Earthquake Tsunami Joint Survey Group [2] analysed nationwide field measurements of inundation height, run-up height, and inundation distance following the 2011 Tohoku tsunami. The observed response varied strongly between flat coastal plains and the steep ria coast of San-riku, demonstrating that regional and bay-scale geometry materially altered the transformation from offshore tsunami elevation to onshore inundation and run-up.

The paper is retained as observational evidence for nearshore transformation and inundation behaviour. It does not provide sufficient dynamic information to determine local velocity, momentum flux, impact pressure, or complete time histories.

5.5 Governing Relations and Quantitative Definitions

The paper used the following empirical linear relation to describe bay-scale variation in measured inundation height:

$$H(x) = ax + b \quad (5.1)$$

where x is distance from the bay mouth, $H(x)$ is the measured inundation height, and a and b are fitted coefficients. The coefficient a was interpreted as an empirical amplification or attenuation indicator.

This relation shall not be adopted as a governing physical model. It is a descriptive fit whose value depends on the selected observations, bay definition, and fitting procedure.

5.6 Physical Mechanisms

The survey indicates that nearshore tsunami behaviour was controlled by the interaction between incident wave conditions and local bathymetry, topography, bay orientation, continental-shelf geometry, river channels, coastal structures, diffraction, and macro-scale roughness.

Steep ria coasts generally converted the incident tsunami into large inundation and run-up elevations over comparatively short inland distances. The flatter Sendai Plain exhibited lower regional elevations but substantially greater inundation distances. Rivers provided preferential pathways for extended inland propagation.

5.7 Characteristic Spatial and Temporal Scales

Maximum run-up of approximately 40.0 m was reported at Ofunato. Run-up exceeding 30 m extended across approximately 180 km of the Sanriku coast, while run-up exceeding 10 m extended across approximately 530 km. On the Sendai Plain, inundation extended up to approximately 5 km inland in the regional analysis, with longer propagation along some river channels. More than 1,000 inundation and run-up observations were available within this region.

5.8 Relevant Observables

The principal nearshore observables identified by the survey are:

- maximum shoreline tsunami elevation;
- maximum local inundation height;
- maximum run-up elevation;
- inundation distance;

-
- along-coast variation in maximum elevation;
 - cross-shore decay or amplification of water level;
 - bay-scale amplification or attenuation.

These observables describe maximum spatial response. They do not describe the complete dynamic state required to determine hydrodynamic loading.

5.9 Controlling Dimensionless Parameters

5.10 Physical Regimes and Transition Conditions

At least two distinct coastal-response regimes shall be represented:

1. flat coastal-plain inundation, characterised by extensive inland propagation and gradual reduction in water level;
2. steep ria-coast run-up, characterised by large elevation, bay-scale amplification or attenuation, and shorter general inland penetration.

River-controlled inundation forms an additional local regime in which the propagation distance may substantially exceed that of the surrounding land surface.

5.11 Implications for Model Validity

A model validated only against offshore or shoreline elevation cannot be assumed to reproduce inland inundation and run-up correctly. Validation must include spatially resolved onshore observables and must use topographic and bathymetric resolution appropriate to the selected coastal regime.

The observed variation between neighbouring bays demonstrates that a coarse regional model may reproduce broad propagation while failing to capture local amplification. The local barrier-interaction model must therefore inherit boundary data from the regional model while resolving the selected bay, shoreline, and structure at materially finer resolution.

5.12 Candidate Approaches

5.13 Critical Comparison

5.14 Assumptions and Applicability

5.15 Limitations and Failure Conditions

The survey records maximum elevations and spatial limits rather than time-dependent velocity, discharge, pressure, or momentum. It cannot, without supplementary evidence, validate the quantities required for barrier-impact loading or structural analysis.

The comparison between Otsuchi and Kamaishi Bays indicates a reduction in measured height near the Kamaishi offshore barrier, but the observation does not isolate the barrier from differences in bathymetry, bay geometry, wave incidence, diffraction, and local structures. The paper shall therefore not be used alone to quantify barrier efficiency.

5.16 Project-Specific Conclusions

The nearshore model shall distinguish coastal-plain inundation from ria-bay run-up and shall preserve rivers, bay geometry, and relevant coastal structures where they materially affect the selected benchmark.

The final case-study location shall not be selected solely from the largest measured run-up. Selection shall consider observational density, availability of dynamic data, bathymetric and topographic resolution, presence of a defensible barrier configuration, and compatibility with the proposed regional-to-local model coupling.

5.17 Selected Approach and Rationale

For the regional model, nearshore transformation and inundation shall be represented by the depth-averaged NLSWE with topography-driven wetting–drying and positivity controls. This establishes the regional physical/numerical requirement; it does not replace local 3D modelling where structure-scale pressure, overtopping or FSI is required.

5.18 Unresolved Decisions

The final coastal regime, case-study site, local bathymetry/topography resolution, defence representation and validation observations remain open until the geospatial/data workstream is locked.

5.19 Requirements and Handoffs

The environmental-data workstream shall identify a geographically bounded set of inundation-height, run-up-height, and inundation-distance observations with complete coordinate and datum metadata.

The numerical-method workstream shall provide wetting–drying, positivity preservation, and sufficiently resolved shoreline treatment for the selected coastal regime. The verification workstream shall use separate spatial error measures for water elevation and inundation extent rather than one combined scalar comparison.

5.20 Deliverable Completion Record

6 RES-PHY-BAR — Barrier–Flow Interaction and Energy Dissipation

6.1 Purpose and Scope

This Deliverable links local barrier-flow mechanisms to the physical outputs extracted from the local three-dimensional URANS–VOF model. It defines what is meant by comparative hydrodynamic barrier performance before structural failure or final certification are considered. The active geometry scope is fixed-rigid comparison of wall-type and obstacle/dissipating/array-type barriers.

6.2 Research Questions

6.3 Terminology and Definitions

6.4 Literature Evidence

Kozelkov et al. [3] supports the claim that turbulence can materially affect run-up, overtopping and barrier loading in Navier–Stokes tsunami simulations. del Jesus, Lara, and Losada [4] supports deferring unresolved porous-barrier physics to a separate activation-gated model. Source roles: ‘3D-SST’, ‘3D-RANSVOF’ and ‘3D-DEFERRED’. Project conclusion: the baseline comparison shall use resolved hydrodynamic outputs for impermeable or explicitly resolved barrier geometries, while unresolved porous dissipation is not claimed unless a porous model is activated.

6.5 Governing Relations and Quantitative Definitions

Overtopping volume during a loading interval is:

$$V_{\text{ot}} = \int_{t_0}^{t_1} \int_{\Gamma_{\text{crest}}} \max(\mathbf{u} \cdot \mathbf{n}_{\text{crest}}, 0) \alpha \, d\Gamma \, dt \quad (6.1)$$

A transmission measure may be defined from incident and transmitted energy-like wave measures:

$$K_{\text{T}} = \frac{E_{\text{trans}}}{E_{\text{inc}} + \epsilon_E} \quad (6.2)$$

Reflection and apparent dissipation measures shall be reported with the same incident-energy normalisation:

$$K_{\text{R}} = \frac{E_{\text{refl}}}{E_{\text{inc}} + \epsilon_E} \quad (6.3)$$

$$D_E = 1 - K_{\text{T}} - K_{\text{R}} - \frac{E_{\text{ot}}}{E_{\text{inc}} + \epsilon_E} \quad (6.4)$$

Decision role: these equations define comparison diagnostics for energy redirection and loss, not a closed mechanical energy proof. Supporting evidence: del Jesus, Lara, and Losada [4] for barrier transmission/reflection distinctions and Kozelkov et al. [3] for turbulence-sensitive run-up and overtopping. Limitation: final energy definitions must be compatible with available validation measurements and numerical outputs.

6.6 Physical Mechanisms

The barrier-flow mechanisms of interest are incident-wave run-up, reflection, overtopping, flow separation, wake formation, local recirculation, transmitted wave reduction, pressure loading and turbulent dissipation. For the baseline impermeable model, energy dissipation is inferred from resolved hydrodynamic changes and turbulence modelling rather than unresolved porous resistance.

Wall-type barriers primarily test reflection, run-up, overtopping, crest overflow and loading. Obstacle, dissipating and array barriers primarily test blockage, gap jets, separation, wake interaction, transmission reduction and downstream momentum redistribution. Source role: this distinction connects RES-GEO classes to measurable flow mechanisms. Supporting citations: del Jesus, Lara, and Losada [4] and Watanabe et al. [5]. Downstream handoff: RES-GEO-MET and RES-VER-MET shall preserve class-specific energy and loading metrics.

6.7 Characteristic Spatial and Temporal Scales

6.8 Relevant Observables

Relevant observables are maximum run-up, overtopping volume or discharge, transmitted wave height or energy measure, reflected-wave measure, downstream maximum depth and velocity, peak pressure, force, impulse, moment and wake/recirculation indicators.

6.9 Controlling Dimensionless Parameters

6.10 Physical Regimes and Transition Conditions

6.11 Implications for Model Validity

6.12 Candidate Approaches

6.13 Critical Comparison

6.14 Assumptions and Applicability

6.15 Limitations and Failure Conditions

6.16 Project-Specific Conclusions

6.17 Selected Approach and Rationale

The selected barrier-performance approach is comparative and hydrodynamic: candidate defences are compared using run-up, overtopping, transmission, reflected energy, pressure, force, impulse, moment and downstream-flow metrics extracted from accepted local 3D simulations. The word accepted here refers to timestep/run acceptance and later verification status; this Deliverable does not claim that validation has been executed.

6.18 Unresolved Decisions

Open decisions are the final energy diagnostic, normalisation of transmission/reflection measures, accepted overtopping threshold, porous-extension activation gate and weighting of hydrodynamic metrics for optimisation.

6.19 Requirements and Handoffs

‘RES-NUM-SPEC’ shall extract these metrics after timestep acceptance. ‘RES-OPT-OBJ’ shall decide how hydrodynamic outputs become objective functions. ‘RES-ML-REP’ shall define which metrics become surrogate outputs, and ‘RES-VER-MET’ shall define validation and acceptance criteria.

6.20 Deliverable Completion Record

7 RES-PHY-LOD — Hydrodynamic Loading and Impact Physics

7.1 Purpose and Scope

This Deliverable defines the hydrodynamic loading observables that the local three-dimensional URANS–VOF model must resolve for barrier assessment and later structural handoff. The active baseline assumes a fixed rigid barrier; structural deformation, material damage, scour and two-way FSI are stretch extensions after hydrodynamic load extraction is stable.

7.2 Research Questions

7.3 Terminology and Definitions

7.4 Literature Evidence

Qin et al. [6] is retained as evidence that local three-dimensional modelling enables direct structural-force extraction from pressure and shear traction. Watanabe et al. [5] is retained as evidence that local force, pressure and velocity validation must be assessed separately from water-level agreement. Source roles: ‘3D-RANSVOF’ and ‘3D-VALIDATION’. Project conclusion: the local model shall output complete pressure and traction histories, not only empirical drag coefficients. Limitation: these quantities remain hydrodynamic loads until structural response or FSI is activated.

7.5 Governing Relations and Quantitative Definitions

The pressure and viscous contributions to traction on a barrier surface are:

$$\mathbf{t}_p = -p\mathbf{n} \quad (7.1)$$

$$\mathbf{t}_\tau = \boldsymbol{\tau}\mathbf{n} \quad (7.2)$$

$$\mathbf{t} = \mathbf{t}_p + \mathbf{t}_\tau \quad (7.3)$$

The loading quantities of interest are:

$$\mathbf{F}(t) = \int_{\Gamma_B} \mathbf{t} d\Gamma \quad (7.4)$$

$$\mathbf{M}(t) = \int_{\Gamma_B} (\mathbf{x} - \mathbf{x}_0) \times \mathbf{t} d\Gamma \quad (7.5)$$

$$\mathbf{J} = \int_{t_0}^{t_1} \mathbf{F}(t) dt \quad (7.6)$$

7.6 Physical Mechanisms

The local impact model shall represent hydrostatic loading, nonhydrostatic impact pressure, bore impact, reflected and transmitted wave loading, overtopping-induced loading, shear traction, wake effects and pressure-duration effects. Structural response is not inferred from these hydrodynamic quantities unless ‘RES-STR’ activates a structural model.

7.7 Characteristic Spatial and Temporal Scales

7.8 Relevant Observables

Required local loading observables are pressure histories, pressure spatial distributions, shear traction, resultant force, force impulse, overturning moment, torsional moment, centre-of-pressure migration, impact duration and repeat-wave cumulative loading.

Each loading observable shall carry the identifiers `corridor_id`, `geometry_id`, `patch_id`, local coordinate frame, integration interval and accepted-timestep status. Source role: Qin et al. [6] supports direct force extraction from local 3D fields; Watanabe et al. [5] supports reporting pressure, velocity and force separately. Downstream handoff: RES-STR-LOD shall consume these histories as hydrodynamic loads, not as proof of material response.

7.9 Controlling Dimensionless Parameters

7.10 Physical Regimes and Transition Conditions

7.11 Implications for Model Validity

7.12 Candidate Approaches

7.13 Critical Comparison

7.14 Assumptions and Applicability

7.15 Limitations and Failure Conditions

7.16 Project-Specific Conclusions

7.17 Selected Approach and Rationale

The selected loading approach is direct traction integration over the resolved barrier surface. Empirical drag-only loading may be retained for screening comparisons or sanity checks but shall not replace the local pressure and shear extraction used for validation and structural handoff.

7.18 Unresolved Decisions

7.19 Requirements and Handoffs

‘RES-NUM-SPEC’ shall extract the loading metrics after each accepted local timestep. ‘RES-VER-MET’ shall define acceptance criteria for pressure, force, impulse and moment. ‘RES-STR-LOD’ shall decide how accepted hydrodynamic tractions are transferred into structural load cases. Software shall emit pressure, shear, force, moment and impulse histories in the structured output schema before any structural damage or FSI extension is attempted.

7.20 Deliverable Completion Record

8 RES-PHY-SCL — Dimensional Analysis, Scaling and Flow Regimes

8.1 Purpose and Scope

This Deliverable records the dimensional and regime arguments that support the local three-dimensional URANS–VOF tsunami–barrier model. It links the high-Reynolds-number turbulence assumption to the physical outputs required by the project.

8.2 Research Questions

8.3 Terminology and Definitions

8.4 Literature Evidence

Qin et al. [6] is retained as evidence that local tsunami impact can require three-dimensional velocity and nonhydrostatic pressure resolution. Kozelkov et al. [3] is retained as evidence that turbulence effects become material during breaking, run-up, overtopping and barrier loading. Source roles: ‘3D-RANSVOF’ and ‘3D-SST’. Project conclusion: the local model shall be treated as a high-Reynolds-number turbulent flow model rather than a laminar diagnostic calculation. Limitation: dimensionless regime arguments do not replace validation of pressure, force and overtopping outputs.

8.5 Governing Relations and Quantitative Definitions

8.6 Physical Mechanisms

8.7 Characteristic Spatial and Temporal Scales

Characteristic scales shall be recorded separately for the regional event, the corridor extract and the local barrier domain:

- regional wavelength, source footprint and propagation time from the Tōhoku source to the target coast;
- corridor width W , offshore/pre-impact length L_{pre} , inland length L_{inland} and sponge/relaxation widths;

-
- local flow depth h , representative velocity U , barrier height, barrier width, gap spacing, crest length and wake length.

The corridor dimensions are physical and numerical scale choices. They shall be derived from source footprint, severe-impact observations, defence geometry, bathymetry/topography coverage and relaxation-zone requirements, not from visual convenience.

8.8 Relevant Observables

8.9 Controlling Dimensionless Parameters

The local flow regime shall be characterised using Reynolds, Froude and Weber numbers:

$$\text{Re} = \frac{UL}{\nu_w} \quad (8.1)$$

$$\text{Fr} = \frac{U}{\sqrt{gh}} \quad (8.2)$$

$$\text{We} = \frac{\rho_w U^2 L}{\sigma} \quad (8.3)$$

Here U is a representative incident or local velocity scale, L is a local barrier or flow-depth scale, h is local water depth, ν_w is water kinematic viscosity and σ is surface tension.

8.10 Physical Regimes and Transition Conditions

The baseline local regime is turbulent, gravity-dominated, free-surface flow with large inertial loading. High Reynolds number supports use of a turbulence closure. Froude number identifies supercritical or near-critical run-up, overtopping and bore-like conditions. Large Weber number supports treating surface tension as a minor tsunami-scale force except in small entrained droplets or aeration details, which are outside the baseline.

The Kamaishi and Sendai corridors represent different physical regimes within the same event. Kamaishi is retained for ria/bay and defence- interaction scaling; Sendai is retained for coastal-plain inundation scaling. Source role: supports active two-corridor comparison. Supporting sources: Mori, Takahashi, and The 2011 Tohoku Earthquake Tsunami Joint Survey Group [2] for contrasting Sanriku and Sendai inundation/run-up behaviour, and U.S. Geological Survey [7] for the event source point used in trajectory construction. Limitation: final corridor bearings and dimensions remain GIS/data products owned by ‘RES-DAT-SCN’.

8.11 Implications for Model Validity

The high-Reynolds-number regime supports URANS $k-\omega$ SST as the operational local model and LES as a higher-fidelity comparator. The same regime also prevents free-surface height alone from validating the model: velocity, pressure, force, impulse, overtopping and wake quantities must be checked separately.

The corridor scale also constrains the validity of one-way replay. If reflection, bay resonance, barrier blocking or coastal confinement propagates materially back through the local inlet, the simultaneous coupling option in ‘RES-MOD-MUL’ shall be tested.

8.12 Candidate Approaches

8.13 Critical Comparison

8.14 Assumptions and Applicability

8.15 Limitations and Failure Conditions

8.16 Project-Specific Conclusions

8.17 Selected Approach and Rationale

The selected regime interpretation is:

local tsunami–barrier interaction is treated as high-Reynolds-number turbulent, gravity-dominated, three-dimensional free-surface flow requiring URANS turbulence closure for operational screening

The selected corridor interpretation is:

use constant-width baseline corridors for Kamaishi and Sendai until GIS evidence shows that a physical bay, coastline, river or bathymetric constraint requires a non-rectangular footprint

8.18 Unresolved Decisions

8.19 Requirements and Handoffs

‘RES-MOD-SRC’ shall retain the turbulence closure. ‘RES-NUM-SPEC’ shall solve the turbulence variables after pressure correction. ‘RES-VER-MET’ shall require validation metrics beyond free-surface agreement.

‘RES-DAT-SCN’ shall provide the corridor dimensions and final bearings. ‘RES-VER-TST’ shall test whether the corridor and interface placement contaminate run-up, overtopping, force, moment or energy metrics.

8.20 Deliverable Completion Record

9 RES-PHY-SYN — Integrated Physical-Phenomena Model

9.1 Purpose and Scope

9.2 Research Questions

9.3 Terminology and Definitions

9.4 Literature Evidence

9.5 Governing Relations and Quantitative Definitions

9.6 Physical Mechanisms

The regional 2D NLSWE model represents long-wave tsunami generation, propagation, refraction over bathymetry, nearshore transformation, run-up and inundation where depth-averaged hydrostatic assumptions are acceptable for the selected outputs. The bed is a positive-upward scalar field and may move dynamically during earthquake forcing.

The regional model does not resolve vertical velocity profiles, nonhydrostatic impact pressure, air–water interface topology, local breaking jets around a structure, structural deformation or FSI. Those mechanisms belong to local 3D/interface/FSI workstreams.

9.7 Characteristic Spatial and Temporal Scales

9.8 Relevant Observables

9.9 Controlling Dimensionless Parameters

9.10 Physical Regimes and Transition Conditions

9.11 Implications for Model Validity

The regional mathematical model is established at research-specification level, but physical credibility still depends on domain selection, bathymetry/topography quality, source-data quality, wet/dry validation and comparison against observations. Agreement in offshore propagation does not by itself validate local structure loading.

9.12 Candidate Approaches

9.13 Critical Comparison

9.14 Assumptions and Applicability

9.15 Limitations and Failure Conditions

9.16 Project-Specific Conclusions

9.17 Selected Approach and Rationale

The selected physical regional baseline is a depth-averaged, hydrostatic, horizontal two-dimensional long-wave model. Optional dispersive or Coriolis sensitivities may be assessed later if the selected domain or validation evidence requires them.

9.18 Unresolved Decisions

Open physical work is final case-study/domain selection, final source and terrain datasets, validation of propagation and inundation, and handoff conditions for local 3D/interface modelling.

9.19 Requirements and Handoffs

9.20 Deliverable Completion Record

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